

Reinforcement or Reflection? College Student Preferences for Gamification Mechanics in Digital Stress-Management Systems

ANONYMOUS AUTHOR(S)

Digital mental health (DMH) applications are increasingly used to support stress management among college students. Many platforms incorporate gamification design to encourage sustained engagement. While these mechanics are designed to reinforce habit formation, it remains unclear whether they align with student preferences in stress contexts. We conducted a cross-sectional survey study ($N = 38$) examining how university students rate the desirability of common gamification mechanics in DMH applications. Participants evaluated narrative-driven, competitive, social, immersive, and progress-based features using Likert-scale ratings framed around experiences of stress. The survey also assessed current self-care practices and DMH app usage. Descriptive findings indicate that narrative and personalization-oriented mechanics were rated more favorably than competitive or externally comparative features. Points and leaderboard systems received the lowest desirability ratings in stress-related contexts. Students reported greater alignment with practices such as taking breaks, goal setting, and relational support. These results highlight a potential misalignment between reinforcement-based gamification strategies and student preferences during periods of stress. By examining gamification at the level of individual mechanics, this study contributes empirical insight on how interaction design choices may differentially support stress-management experiences among college students.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; *Interaction design*; • **Applied computing** → *Consumer health*; Health informatics.

Additional Key Words and Phrases: Gamification, Digital Mental Health, Stress Management, Interaction Design, College Students, Narrative Systems

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1 Introduction

Digital mental health (DMH) interventions have demonstrated measurable improvements in stress, resilience, and psychosocial well-being among adolescents and young adults [1, 8, 16]. However, these benefits depend heavily on sustained engagement, which remains a persistent challenge across digital health systems [7, 14]. As DMH platforms proliferate, the design question increasingly shifts from whether these systems *can* improve mental health to how they should be designed to support meaningful, ongoing use in everyday contexts.

This question is particularly salient for college students, who consistently report elevated stress related to academic pressure, financial strain, and time management [1, 2]. Although many students utilize mobile and web-based mental health tools, engagement patterns are often inconsistent [7, 14]. DMH applications therefore occupy a central but unstable position within student stress-management ecosystems.

To address engagement challenges, many DMH systems incorporate *gamification*, defined as the use of game design elements in non-game contexts [6]. Common mechanics include points, streaks, badges, leaderboards, and progress tracking. These structures are typically designed to reinforce habits and encourage return behavior [14]. While such mechanics may increase usage frequency, interaction design choices are not neutral. From an affordance perspective,

interface features shape how users interpret and act within systems [4, 11]. In stress-management contexts, gamified mechanics may frame self-care as achievement-oriented, competitive, or performance-based.

Stress, however, is not solely physiological; it is also interpretive. How individuals appraise distress influences coping and regulation [11]. Mechanics that emphasize comparison and measurable progress may inadvertently reproduce evaluative pressures common in academic environments. In contrast, narrative-driven and personalization-oriented systems may support reflection, agency, and identity expression [3, 9].

These observations suggest a design tension: reinforcement-based gamification strategies optimized for compliance may not fully align with the interpretive needs of stressed college students. Despite widespread adoption of gamification in DMH systems, limited empirical work examines how specific mechanics are perceived at the feature level within stress contexts. Prior research often evaluates overall engagement or clinical outcomes rather than isolating perceptions of individual interaction mechanics [7, 14].

To address this gap, we conducted a cross-sectional survey study ($N = 38$) examining how university students rate the desirability of different gamification mechanics in stress-management applications. Participants evaluated narrative-driven, competitive, social, and progress-based mechanics using Likert-scale ratings framed around experiences of stress.

We investigate the following research questions:

- **RQ1:** How do university students rate different gamification mechanics in the context of digital mental health applications?
- **RQ2:** Which categories of mechanics are perceived as most desirable when students are experiencing stress?

Our descriptive findings indicate that narrative and personalization-oriented mechanics were rated more favorably than competitive or socially comparative features. Competitive elements—frequently emphasized in commercial wellness platforms—received the lowest desirability ratings in stress contexts. These results highlight a potential misalignment between dominant reinforcement-based gamification patterns and student preferences during periods of stress.

By examining gamification at the level of individual mechanics, this study contributes empirical insight to Interactive Health research on how specific interaction features may differentially support—or undermine—stress-management experiences among college students.

2 Related Work

2.1 Gamification in Interaction Design

Gamification is defined as the use of game design elements in non-game contexts [6]. Within HCI, gamification has been widely applied to motivate engagement and support behavior change through mechanics such as points, leaderboards, streaks, badges, progress tracking, and narrative systems [14].

Contemporary scholarship increasingly analyzes these mechanics as distinct interaction affordances rather than treating gamification as a singular construct [4, 10]. From motivational and affordance-based perspectives, mechanics may emphasize competence (e.g., achievement systems), autonomy (e.g., exploratory progression), or social comparison (e.g., leaderboards). Narrative and immersive elements have been associated with identity exploration and interpretive engagement [3, 9], whereas competitive mechanics foreground measurable performance and rank-based reinforcement.

This feature-level perspective suggests that interaction mechanics are not interchangeable. In health-related contexts, different mechanics may shape not only engagement patterns but also how users interpret progress, competence, and distress [11].

2.2 Gamification in Digital Mental Health

Digital mental health (DMH) applications frequently integrate gamification to encourage adherence to stress-management and self-care practices. Mobile mindfulness and resilience interventions commonly employ streak systems, goal tracking, aesthetic rewards, and structured progression to reinforce continued use [8, 12].

While such systems can improve stress-related outcomes [1], engagement remains inconsistent [12]. Many commercial DMH platforms rely heavily on reinforcement-oriented mechanics emphasizing consistency, metrics, and comparison [14]. In contrast, some systems incorporate narrative or immersive elements inspired by commercial games to support reflection or experiential engagement [3, 9].

Given this variation in design approaches, more research is needed to understand how distinct gamification mechanics are perceived in stress contexts.

2.3 Feature-Level Gaps in Stress Contexts

Much existing research evaluates overall system engagement or clinical outcomes rather than comparing perceptions of individual mechanics [7, 14]. In stress-management settings—particularly among college students who experience elevated academic pressure—interaction framing may be especially consequential [1, 7]. Competitive and socially comparative mechanics may reproduce evaluative pressures, whereas narrative and personalization-oriented structures may support alternative interpretive frames [9, 11].

A feature-level examination of perceived desirability can therefore inform more context-sensitive interaction design in DMH systems. In this study, we empirically examine how university students evaluate narrative-driven, competitive, social, and progress-based mechanics within stress-management applications.

3 Method

3.1 Study Design

We conducted a cross-sectional online survey using Qualtrics to examine university students' preferences for gamification mechanics in DMH stress-management contexts [14]. The instrument was designed to evaluate perceptions of individual interaction mechanics commonly implemented in commercial DMH applications rather than to assess clinical symptom severity.

Survey items were informed by prior literature on gamification, affordances, and digital mental health engagement [3, 6, 9, 14]. Items were adapted to reflect mechanics frequently encountered in mobile wellness platforms while remaining contextually framed around experiences of stress.

The study protocol was approved by the University Institutional Review Board (IRB). Participants provided informed consent electronically prior to beginning the survey.

3.2 Participants and Recruitment

Participants were recruited from a midsized university through campus flyers and in-person outreach at student group activities. Eligibility criteria required participants to be currently enrolled students and at least 18 years of age.

A total of $N = 104$ students initiated the survey. The survey was voluntary, uncompensated, and could be exited at any time. For analytic consistency, responses were included only if participants completed the full gamification preference section. After excluding partial completions, the final analytic sample consisted of $N = 38$ completed surveys.

Item-level missingness varied slightly across questions (range: $N = 35$ – $N = 38$), and analyses were conducted using available responses per item.

3.3 Survey Instrument

The questionnaire consisted of multiple matrix-style and single-response items organized into five sections: (1) demographics, (2) mental wellbeing perceptions, (3) gamification mechanic preferences, (4) social support preferences, and (5) technology and engagement preferences.

Although several items were presented within matrix blocks, each row functioned as an individual analyzable item. In total, the instrument included demographic questions, one self-care app usage item (with optional explanation), multi-item stressor ratings, multi-item coping activity ratings, multi-item gamification preference ratings, mood-related mechanic evaluations, progress visualization ratings, social support items, and technology feature preference items. Full item wording is provided in Appendix A.

Unless otherwise specified, responses were measured using 5-point Likert-type scales. Scale anchors varied by section (e.g., Never–Always; Not at all useful–Extremely useful; One of the worst–One of the best; Strongly disagree–Strongly agree). The use of 5-point scales was informed by survey methodology guidance in HCI research [5, 13].

3.3.1 Mental Wellbeing Perceptions. Participants rated the frequency with which common domains (e.g., academic pressure, financial concerns, time management, social relationships) contributed to their stress [1]. They also rated the perceived usefulness of common stress-management activities (e.g., meditation, exercise, journaling, counseling, DMH applications) [8, 12]. Participants indicated whether they currently used DMH applications (yes/no) and could optionally describe their experiences.

3.3.2 Gamification Mechanic Preferences. Participants rated the desirability of individual mechanics commonly implemented in DMH applications, including points, leaderboards, progress tracking, personalization features, narrative quests, collaborative activities, immersive elements, collection systems, humor, physical activity mechanics, and problem-solving features [6, 9].

Additional items assessed preferences for mood tracking charts, streak systems, goal progression feedback, and narrative journey representations. All items were explicitly framed within a stress-management context (e.g., “when experiencing stress”).

3.3.3 Social and Technology Preferences. Participants rated their willingness to engage in collaborative self-care activities and receive encouragement from others [10]. They also evaluated preferences for digital stress-relief features such as guided exercises, personalization, community chat, goal reminders, and adaptive or sensor-informed systems [11].

3.3.4 Demographics. Participants reported age (18+ confirmation), gender identity, academic year, and major.

3.4 Data Analysis

Descriptive statistics were computed for all Likert-scale items. Means and standard deviations were calculated for individual mechanics. For category-level analyses, conceptually related items (e.g., narrative elements, competitive mechanics, progress-tracking features) were grouped by averaging constituent item ratings. These groupings were theory-informed and used for exploratory descriptive comparison rather than scale validation.

4 Results

A total of 38 students completed the survey; however, due to item-level missingness, the number of valid responses varied slightly across sections ($N = 35$ – $N = 38$ depending on the item). All analyses are descriptive and reflect Likert-scale ratings (1–5), with higher values indicating stronger endorsement or perceived usefulness.

4.1 RQ1: Preferences for Gamification Mechanics

To address **RQ1**, we examined students' ratings of core gamification mechanics commonly implemented in DMH applications. From our sample, $N = 22$ participants reported current usage of DMH applications, whereas $N = 28$ did not use these tools. One participant expanded on the rationale of not using these DMH applications:

"I guess there's something inhuman about an app that puts me off. Without an in-person connection with a therapist or something, I find it harder to shake the feeling that someone is trying to sell me something. I like the idea of designing games to be relaxing and stuff; like, Journey was an extremely memorable experience for me, probably one of my favorite games ever."

As shown in Table 1, narrative-driven quests received the highest mean rating ($M = 3.50$, $SD = 1.18$), followed by personalization features ($M = 3.42$, $SD = 1.20$). Collaborative activities were rated moderately ($M = 3.13$, $SD = 1.27$). Progress bars and leveling systems fell slightly below the neutral midpoint ($M = 2.97$, $SD = 1.29$). Points and leaderboard-based mechanics received the lowest desirability ratings ($M = 2.15$, $SD = 1.41$).

Table 1. Student ratings of core gamification mechanics in DMH applications (RQ1). Ratings use a 1–5 scale; higher values indicate greater perceived desirability. Ns reflect valid responses per item.

Game Design Mechanic	Mean	SD
Story-driven quests	3.50	1.18
Personalization features	3.42	1.20
Collaborative activities	3.13	1.27
Progress bars / leveling systems	2.97	1.29
Points / leaderboard mechanics	2.15	1.41

Descriptively, mechanics emphasizing narrative structure and customization were rated more favorably than ranking-based systems. Notably, leaderboard mechanics were rated substantially lower than other features. However, not all reward-oriented mechanics were uniformly low; mid-range ratings for collaborative and progression-based features indicate variability within reinforcement-oriented designs.

4.2 RQ2: Category-Level Trends During Stress

To examine broader interaction patterns in line with **RQ2**, we grouped conceptually related items and analyzed category-level trends (see Table 2). Narrative-oriented mechanics (engaging stories and exploration) received the highest grouped mean ($M = 3.46$, $SD = 1.16$), followed by immersive or aesthetic elements such as relaxing visuals and humor ($M = 3.32$, $SD = 1.23$). Social mechanics (cooperative play) were rated moderately ($M = 3.03$, $SD = 1.29$).

Participants who elaborated on their certain types of stress were impeded by their social and perceived expectations. These insights corresponded with the overarching findings from our sample, illustrating the self-perceived internal conflict experienced by the participants and their adaptive responses to it. According to one participant:

"Not being able to meet my own expectation of myself."

Upon expanding on the activities to support wellness, one stated:

"Music, walking, imagine situation where I win over my insecurities."

This reconstruction of thought patterns through imagination aligns with creative expression through immersion in this participant's own self[17].

Competitive/reward-oriented mechanics (e.g., collection systems) were rated near the midpoint ($M = 3.16$, $SD = 1.23$), suggesting neutral to moderately positive perceptions rather than strong endorsement or rejection. In contrast, physical activity mechanics received the lowest ratings overall ($M = 2.13$, $SD = 1.35$). Category-level means were computed by averaging constituent item ratings within each interaction type.

Table 2. Category-level trends for (a) mood/stress support ratings and (b) wellness goal meaningfulness ratings (RQ2). Ratings use a 1–5 Likert scale; higher values indicate stronger endorsement.

Category / Item Group	Mean	SD
<i>Mood/Stress Support</i>		
Narrative (stories + exploration)	3.46	1.16
Immersive/Aesthetic (visuals + humor)	3.32	1.23
Social (cooperative play)	3.03	1.26
Competitive/Reward (collection systems)	3.16	1.23
Physical (physical activity mechanics)	2.13	1.35
<i>Wellness Goal Meaningfulness</i>		
Progress tracking (charts + streaks + metrics)	2.44	1.28
Narrative journey (quest map representation)	2.92	1.22

We also examined perceptions of goal-support mechanisms within wellness applications. Traditional progress-tracking features (e.g., mood charts, streaks, improvement metrics) were rated lower overall ($M = 2.44$, $SD = 1.28$) compared to narrative journey or quest-based representations of progress ($M = 2.92$, $SD = 1.22$). Although both categories clustered near the midpoint, narrative representations of progression were consistently rated more favorably than conventional tracking tools.

4.3 Observed Pattern Across Sections

Across feature-level and category-level analyses, narrative and personalization-oriented mechanics were among the highest-rated features. Leaderboard-based ranking systems were consistently rated lowest. Other reinforcement-oriented mechanics, such as collection systems and progression structures, were evaluated more neutrally. Combined, these emerging patterns suggest that students in this sample may align differently between competitive ranking systems and other forms of reward or progression. While externally comparative mechanics were rated less favorably, immersive and individualized interaction structures were more positively received within stress-management contexts. No causal inferences can be drawn from these findings, but they showcase a disconnect our sample faces within the DMH structures.

5 Discussion

This study examined how university students perceive different gamification mechanics within DMH applications in stress-management contexts. Across both feature-level and category-level analyses (see Tables 1 and 2), a consistent descriptive pattern emerged: narrative-driven and personalization-oriented mechanics were rated more favorably than competitive and externally comparative features.

5.1 Reinforcement-Based Gamification and Stress Contexts

Many commercial DMH platforms rely on reinforcement-oriented mechanics such as streaks, points, leaderboards, and progress metrics [12, 14]. These structures are typically designed to encourage habit formation and return behavior. However, in this sample, competitive mechanics received the lowest desirability ratings, and traditional progress-tracking tools were rated lower than narrative journey representations.

These findings suggest a potential misalignment between reinforcement-centric engagement strategies and how students prefer to engage with stress-management technologies. While competitive systems may support task completion, they may also replicate evaluative dynamics already embedded in academic environments. In stress contexts, mechanics that emphasize comparison or performance may not be experienced as supportive.

5.2 Narrative and Personalization as Alternative Framings

Narrative-driven and personalization-oriented mechanics were rated most favorably. Prior HCI research has linked narrative and immersive systems to reflection, identity exploration, and experiential engagement [3, 9]. From an affordance perspective, interface features shape how users interpret and act within digital systems [4, 11].

In stress-management settings, mechanics emphasizing story progression or customization may frame self-care as personal growth rather than measurable performance. Although this study did not directly assess motivational orientation, the relative pattern of ratings indicates that autonomy-supportive and narrative-oriented mechanics may better align with student preferences during periods of stress.

5.3 Implications for Interactive Health Design

For the Interactive Health community, these findings reinforce the importance of evaluating gamification at the level of individual mechanics rather than treating it as a uniform engagement strategy. Reinforcement-based systems may increase usage frequency, but they do not necessarily translate into perceived support in stress contexts.

Design strategies emerging from this work include:

- Framing wellness activities within narrative or journey-based structures rather than purely metric-based progression [3, 9, 17].
- Incorporating personalization features that support autonomy and identity expression [4, 11, 18].
- Carefully evaluating the use of competitive or comparative mechanics in stress-related applications [14, 18].

Future research should examine how narrative-centered and autonomy-supportive mechanics influence not only engagement patterns, but also perceived stress and coping skill development across diverse student populations [8, 16].

5.4 Limitations

Several limitations should be considered when interpreting these findings. This study employed a cross-sectional, descriptive design with a modest final sample size ($N = 38$) drawn from a single university. Although 104 students

initiated the survey, analyses were limited to participants who completed the full gamification preference section. The voluntary and uncompensated nature of participation may have contributed to attrition, particularly given the time constraints common within student schedules between classes.

The survey relied on self-reported perceptions of feature desirability rather than observed behavioral engagement. Reported preferences may not directly translate into sustained usage patterns within deployed systems. Future research should examine how feature-level perceptions correspond to longitudinal engagement and stress-related outcomes.

Interaction mechanics were grouped conceptually for category-level analyses based on thematic classifications identified in prior gamification and digital mental health literature (see Appendix A). These groupings were theory-informed, and internal consistency was not assessed. Alternative grouping strategies may yield different descriptive patterns.

Survey items were explicitly framed around perceived benefit for alleviating stress and supporting mental wellness. While this contextual framing was intentional, it may have shaped how participants evaluated mechanics relative to more neutral or habit-formation contexts. Preferences may differ in systems designed for fitness, productivity, or general wellbeing rather than acute stress management. Accordingly, findings should be interpreted as exploratory and may not generalize to broader or more diverse student populations.

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A Survey Instrument

Table 3. Full Survey Instrument, including wording with Rating Scales for Each Item

Section Group	Survey Item (as administered)	Response Type	Informed by
Demographics	Age, Gender, Year in College, Major	Age: open-ended Gender: Male, Female, Non-Binary, Prefer Not to Say Year in College: Freshman, Sophomore, Junior, Senior, Graduate Current Major: Open-ended	[13]
Mental Well-Being	Do you currently use any self-care apps?	Yes/No	[7, 14]
Mental Well-Being	Which of the following are your primary sources of stress? Academic pressure Financial concerns Time management Social life and relationships Physical health Family expectations Loneliness or isolation Other (please specify)	Rated on a 1 (Never) to 5 (Always) with one open-ended question	[1, 2]
Mental Well-Being	Which activities do you find the most helpful to maintain your mental wellness? Meditation or mindfulness exercises Physical exercise or sports Journaling or self-reflection Talking to friends or family Counseling or therapy Using mental health apps Taking breaks or vacations Setting personal goals Other (please specify)	Rated on a 1 (Not at all useful) to 5 (Extremely Useful) with one open-ended question	[8, 12, 15, 16]
Gamification	Rate these game mechanics based on personal preference: Point systems and leaderboards Progress bars and levels Personalization Story-driven quests Collaborative activities	Rated on a 1 (One of the worst) to 5 (One of the best)	[6, 14]
Gamification	Do these game elements help improve your mood or reduce stress? Relaxing/calming visuals Engaging stories Creativity Cooperative play Collection (achievements/rewards/items) Exploration Physical activity Problem-solving Humor or lighthearted content Other (please specify)	Rated 1 (Strongly Disagree) to 5 (Strongly Agree) with one open-ended question	[4, 9–11]

Table 3 continued from previous page

Section Group	Survey Item (as administered)	Response Type	Informed by
Gamification	How meaningful are the below game mechanics in helping you keep your wellness goals in a mental wellness app? A weekly or monthly mood chart A calendar marking "streaks" or consistent days of self-care Progress toward personal wellness goals A journey or quest map that reflects personal achievements Other (please specify)	Rated 1 (Not at all important) to 5 (Extremely Important) with one open-ended question	[6, 13, 14, 16]
Gamification	How important do you find features that track and celebrate your progress?	Rated 1 (Not at all important) to 5 (Extremely important).	[6, 14]
Social Support	How likely are you to perform self-care challenges in a team setting where you could work on similar goals?	Rated 1 (Not Likely) to 5 (Very Likely)	[10?]
Social Support	How interested are you in receiving encouragement or motivational messages from others in a group setting?	Rated 1 (Not interested) to 5 (Extremely Interested)	[10?]
Technology and Engagement	Rate your preference on the following game elements in a stress management app? Relaxation exercises Mood tracking Personalized recommendations Community support or chat features Goal-setting and reminders Gamified elements Interactive content Other (please specify)	Rated 1 (Do not prefer) to 5 (Prefer a Great deal). with one open-ended question	[12, 14, 16]
Technology and Engagement	Would you find it helpful if the app could adjust content based on real-time information (e.g., stress level or heart rate) to dynamically change the way you interact with the application?	Rated 1 (Not at all helpful) to 5 (Extremely Helpful)	[5, 13]